Kalyan Pendem

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Skills

Languages: Python, TypeScript, JavaScript
Databases: MongoDB, SQL, PostgreSQL, Neo4j, Redis
Cloud Platforms & Distributed Systems: AWS, GCP, Docker, SQS, Lambda
Full-Stack Technologies: React, Node.js, Express.js, NEXT, REST, GraphQL
Testing & DevOps: Jest, Git, Linux/Unix
Applied AI: RAG, LLM pipelines, LangChain

Work Experience

Backend Engineer (Contract), Confidential Client (Healthcare) — July 2025 – Nov 2025
(Typescript, Node.js, Express, Jest, SQS, CloudWatch)

- Designed an asynchronous SQS-based pipeline to decouple microservices, explicitly trading consistency for throughput — reducing request latency from 8s to <2s while sustaining 500K+ req/hr under burst traffic.
- Led production debugging of auth failures across concurrent requests; redesigned role-based access control and resolved token validation edge cases, improving API latency by 35%.
- Hardened production services by remediating 15+ critical/high Snyk vulnerabilities, introducing automated secret rotation, and enforcing 90% test coverage via CI pipelines.

Software Development Intern (Full-Time), Sense Semiconductor and IT Solutions — Apr 2024 – May 2025
(Typescript, Node.js, Next, Python, GenAI, LangChain, Pinecone)

- Architected a full-stack LMS serving 5,000+ students; identified session persistence as the primary failure risk and introduced Redis-backed session storage to prevent data loss during long-running assessments.
- Built a retrieval-augmented AI tutor, tuned precision vs. recall tradeoffs in the retrieval pipeline, and reduced instructor support tickets by 50%.
- Owned serverless data storage and Lambda-based autoscaling compute pipelines, enabling reliable scale to 500+ concurrent users during peak assessment loads.

Projects

AI-Powered Analysis Platform for CDR/IPDR Data (ReactJS, Python, FastAPI) — CDR-IPDR

- Designed and implemented a forensic data analysis platform that processes large-scale CDR/IPDR datasets into a Neo4j knowledge graph (100K+ records), entity correlation, and investigative query workflows.
- Orchestrated a query-generation pipeline that maps natural-language inputs to safe, bounded Cypher queries (up to 4-hop traversal), enabling reliable investigation workflows on Neo4j graphs.
- Evaluated query correctness using a hybrid approach: rule-based validation (syntax + schema constraints) and LLM-as-judge scoring (factual grounding, query alignment), achieving 90% correctness on a curated benchmark
- Published in IJARSCT — proposed a NL-to-Cypher pipeline for forensic graph traversal on CDR/IPDR datasets.

Sphere – Educational Platform (Typescript, NodeJS, NEXT, PineCone, SQL, RAG) — Maker-or/xs

- Engineered an end-to-end RAG system with LLM-based chat, semantic search and structured file indexing for academic material querying; system served 500+ active students.

Education

Vasireddy Venkatadri Institute Of Technology, B.Tech in Computer Science — Oct 2022 – Apr 2026

- CGPA: 8.5 / 10

Achievements & Community

- Secured 3rd place in a state-level hackathon by developing an agentic knowledge graph system to uncover relational patterns within complex datasets.
- National Finalist (Top 7) in the AI/ML Innovation Hackathon (OpenAI Innovation Category) for developing an educational chatbot at Pharmathon.
- Data Structures & Algorithms: Solved 500+ LeetCode problems, consistently deriving optimal time/space solutions across graph, DP, and tree-based patterns.